

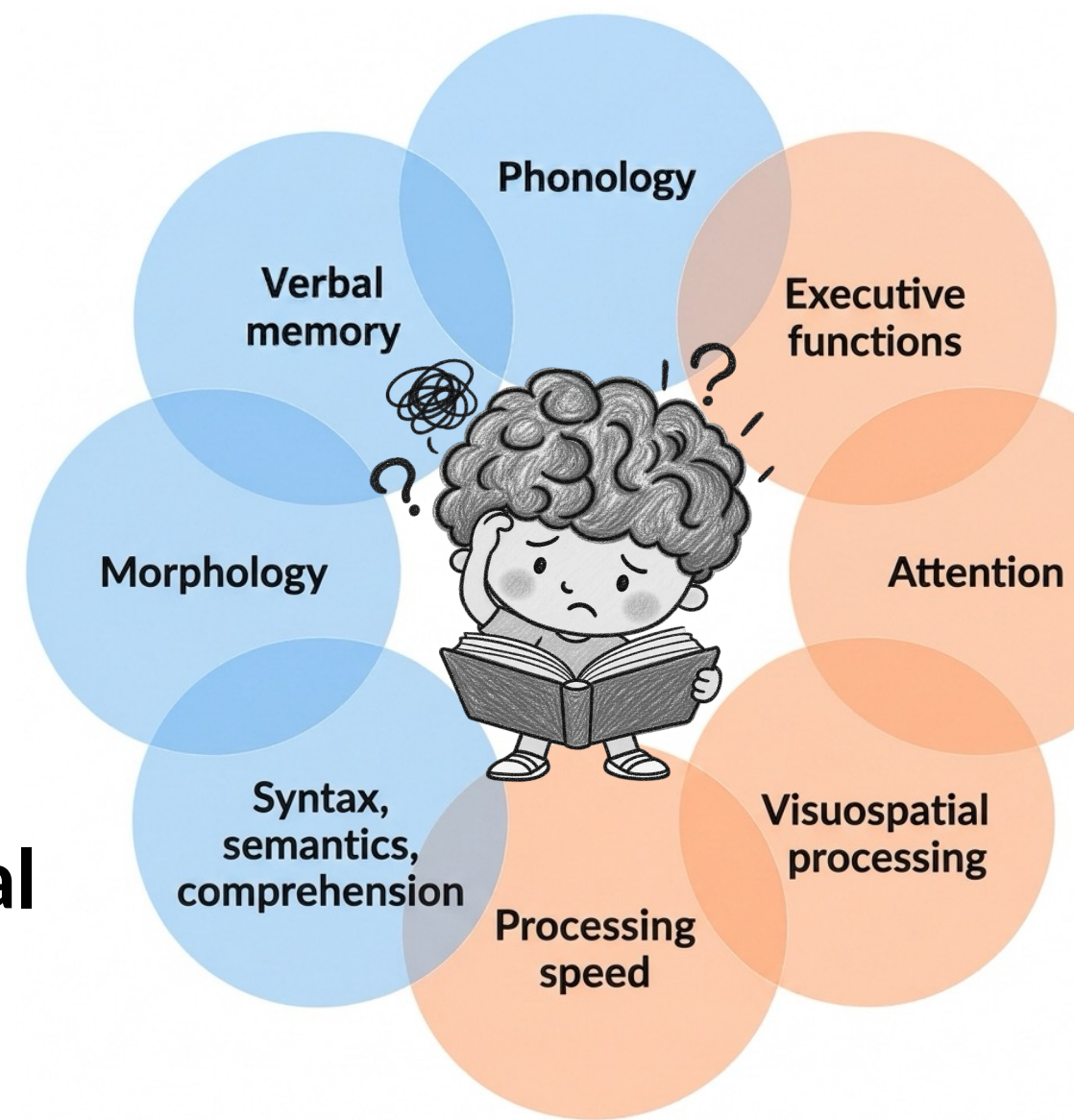
Background

- Developmental dyslexia (DD) is heterogeneous: Similar reading difficulties can arise from distinct cognitive mechanisms.

- Data-driven profiling of **147 extensively phenotyped children with persistent DD** identified two cognitive profiles despite comparable reading difficulties (Pinheiro-Chagas, Kersey et al., 2026):

DD-VSTM

- Phonology
- Verbal short-term memory
- Word retrieval



DD-EF

- Executive functioning
- Processing speed

- The neurobiological basis of cognitive heterogeneity in DD remains unclear, knowledge of which may help better characterize the nature of persistent reading difficulties and develop targeted, biologically informed interventions.

Research Questions

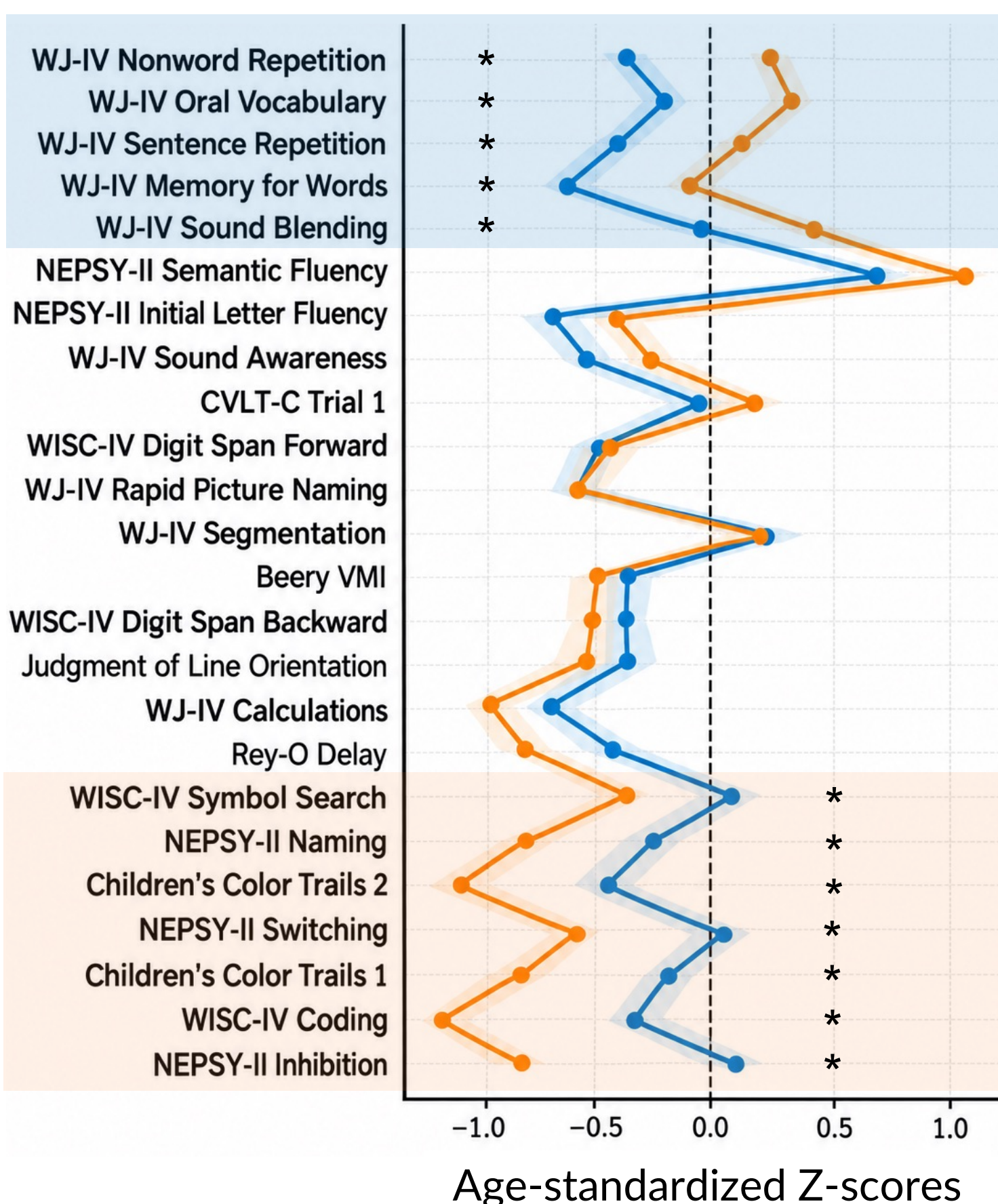
- Do DD-VSTM and DD-EF show distinct patterns of white-matter connectivity relative to neurotypical readers?

DD-VSTM: Left dorsal pathways (AF/SLF) implicated in language
DD-EF: Pathways supporting executive control & attention (SLF, anterior CC, frontostriatal)

- Are there gray-matter regions that can predict each cognitive profile relative to neurotypical readers?

DD-VSTM: Left perisylvian network (IFG, STG/MTG, IPL)
DD-EF: Frontoparietal control network (IFG/MFG, DLPFC)

Participants & Cognitive Profiles

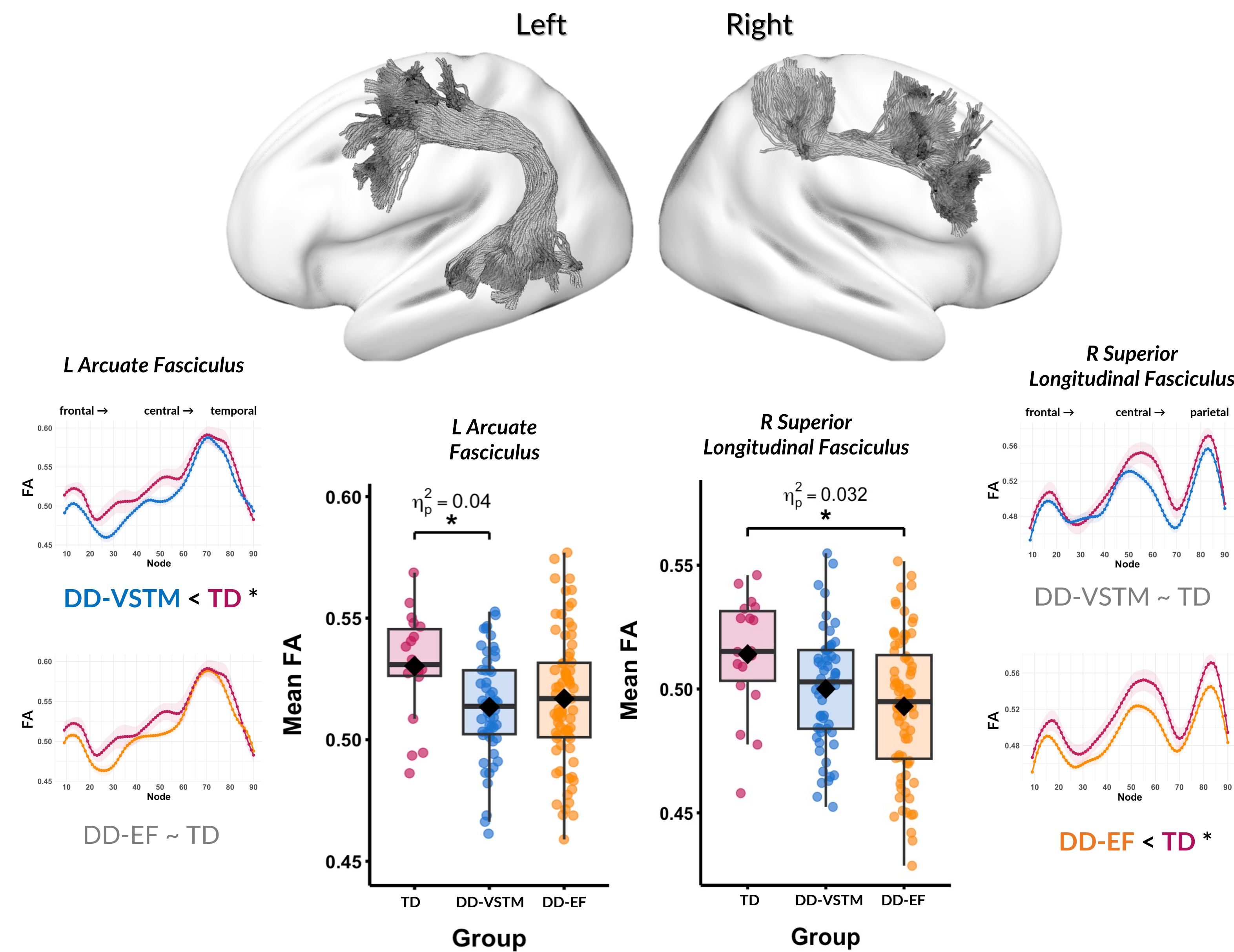


- UCSF Dyslexia Centre DD diagnosis:**
 - ✓ Parent questionnaires
 - ✓ Clinical history
 - ✓ Comprehensive assessment
 - ✓ Multidisciplinary conference
- 24 normed cognitive measures → latent profile analysis
- Within-participant normalization → relative cognitive strengths and weaknesses

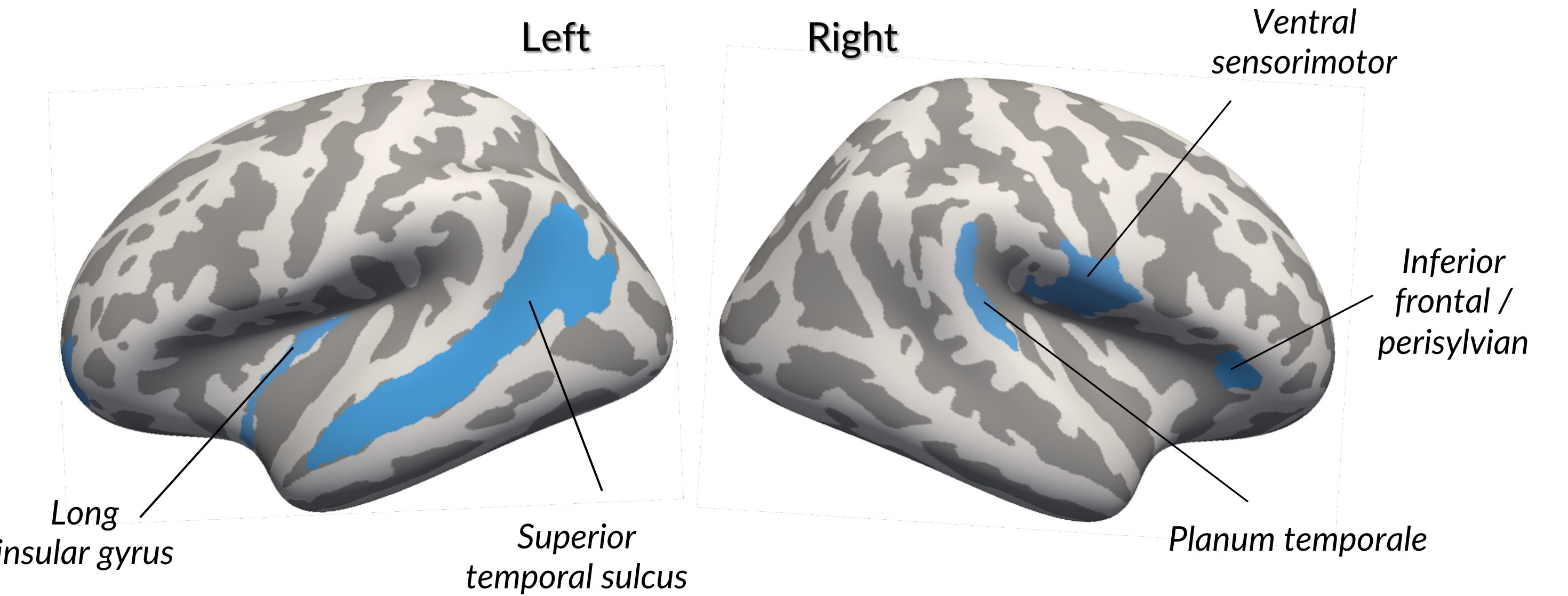
Group (N)	DD-VSTM (58)	DD-EF (76)	N = 20 (Study 1)	N = 29 (Study 2)
Age (SEM)	10.17 (0.23)	9.90 (0.17)	10.75 (0.4)	11.16 (0.3)
Age range	7 - 13 y	7 - 13 y	8 - 14 y	8 - 14 y
Male (%)	53%	58%	30%	35%
R-hand (%)	81%	91%	95%	96%

Results

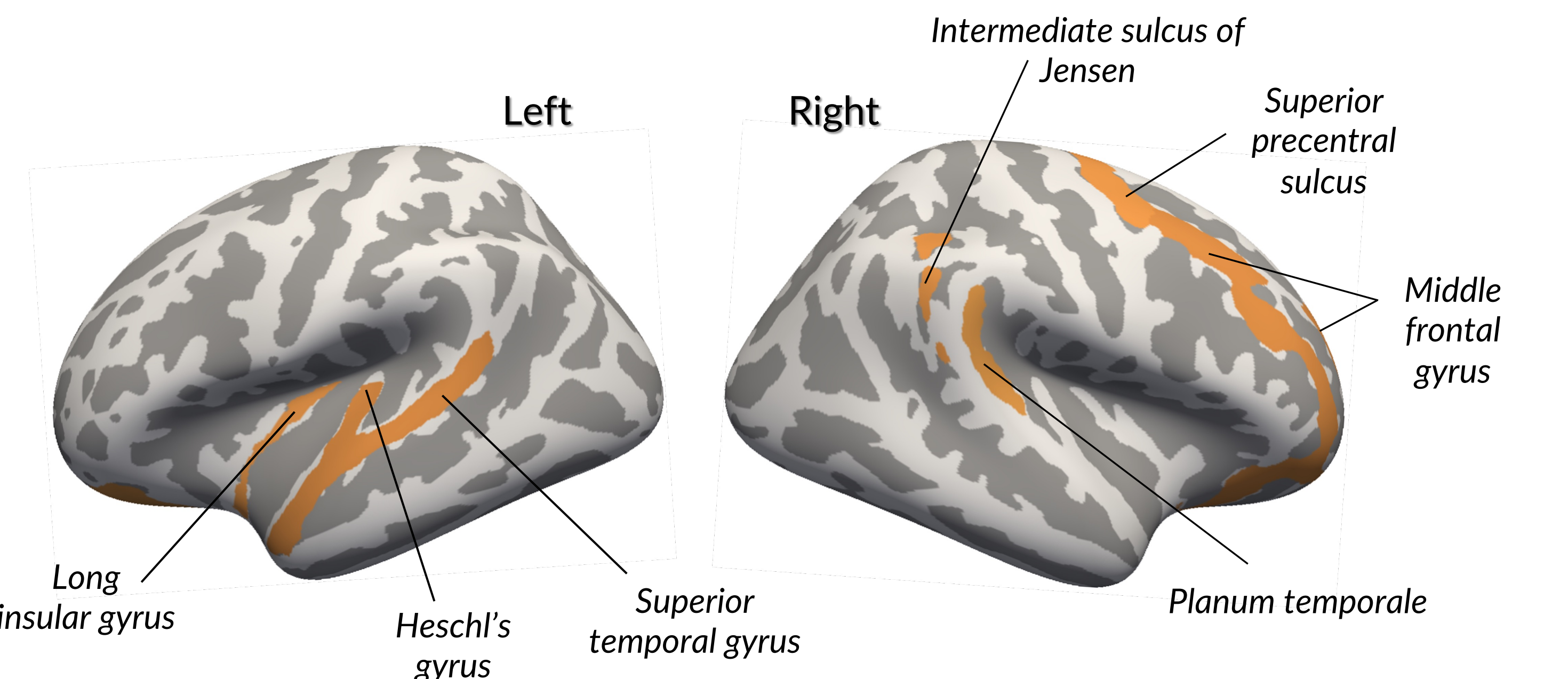
Dyslexia profiles showed modest, profile-specific white-matter differences in two dorsal tracts relative to neurotypical readers



- Lower mean fractional anisotropy in the left arcuate fasciculus in DD-VSTM relative to TD, with group differences along the anterior-to-middle portions of the tract.
- Lower mean fractional anisotropy in the right superior longitudinal fasciculus in DD-EF relative to TD, with group differences observed across much of the tract.



- Predictive cortical volume features distinguishing DD-VSTM from TD included parts of the left and right temporal lobes, alongside insular, ventral sensorimotor, and orbitofrontal regions.



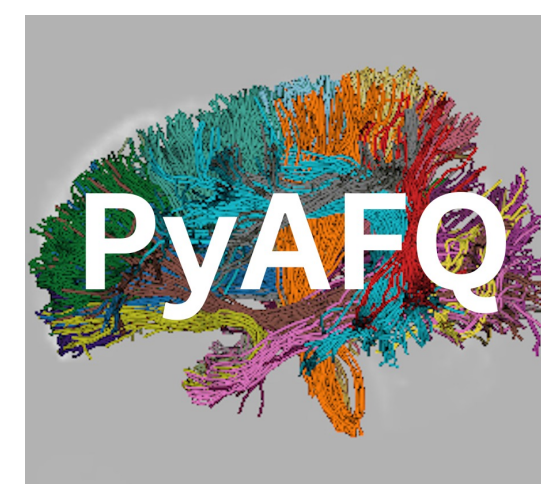
- Predictive cortical volume features distinguishing DD-EF from TD included prominent right frontal and bilateral temporal involvement, as well as insular, precentral, and ventral occipitotemporal regions.

Materials & Methods

White-matter

MRI acquisition & image processing

- 3T Siemens Prisma (64-channel head coil)
- T1w + multi-shell dMRI
- QSIprep:** denoising, susceptibility, eddy-current & motion correction
- Mean framewise displacement (FD) quantified volume-to-volume head motion
- MRtrix3:** whole-brain probabilistic tractography
- pyAFQ:** automated segmentation of a priori white-matter tracts
- FA sampled at 100 nodes/tract
- Mean FA calculated across the middle 80% to minimize endpoint variability



Statistical approach

$$\text{Mean FA}_{\text{tract}} = \beta_0 + \beta_1(\text{Group}) + \beta_2(\text{Age}) + \beta_3(\text{Sex}) + \beta_4(\text{Mean Frame Displacement}) + \epsilon$$

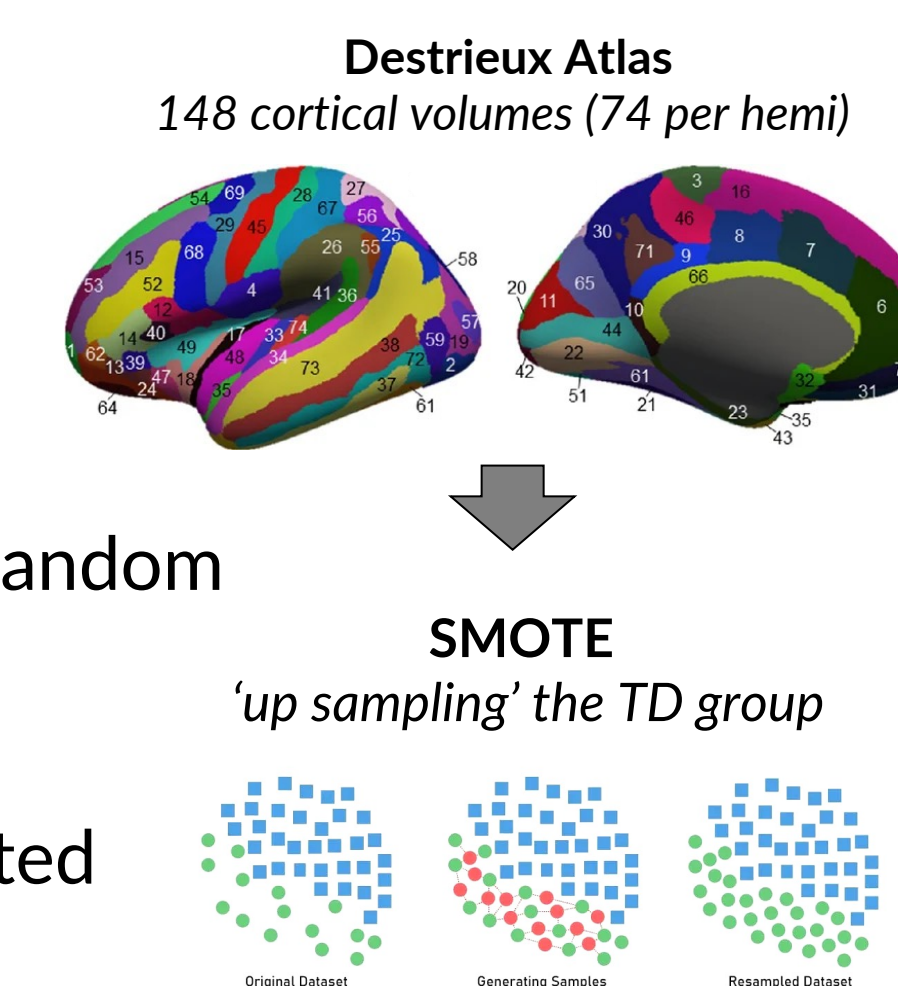
Grey-matter

MRI Acquisition & Image Processing

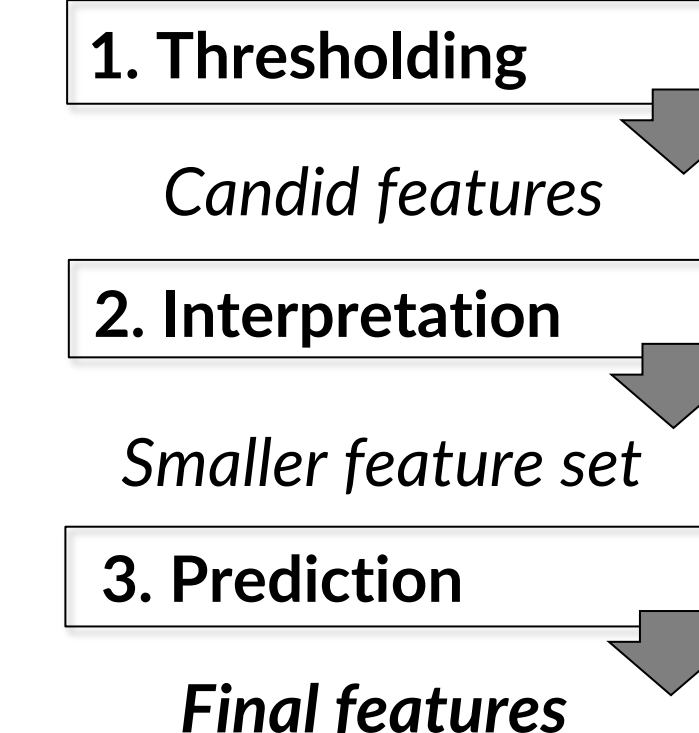
- 3T Siemens Prisma (64-channel head coil)
- MPRAGE sequence

Statistical Approach

- Synthetic Minority Oversampling (SMOTE)
- Variable Selection Using Random Forests (VSURF)
- Data-driven selection of most informative, parcellated cortical volumes



- Random forest variable importance estimated repeatedly, remove noisy variables
- Features ranked by importance, retain a parsimonious explanatory subset
- Remove redundant predictors



Conclusions & Next Steps

- White-matter findings revealed profile-specific differences. DD-VSTM showed lower FA in the left AF, consistent with its role in auditory and phonological processing, whereas DD-EF showed lower FA in the right SLF, a frontoparietal pathway involved in attention and executive control.
- Predictive modelling identified distinct and overlapping cortical patterns. DD-VSTM was distinguished from TD by regions supporting auditory/phonological and speech-sensorimotor processing. DD-EF also involved auditory/phonological regions, but showed greater right-frontal involvement consistent with the involvement of these regions in attention and cognitive control.
- Next steps:** Examine how these neural profiles and their associations with cognitive skills vary across grades to characterize developmental differences in DD. We will also use fixel based analysis to better capture white-matter differences that may be missed due to crossing fibers in the current method of measuring FA.

Scan for References
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